

Module 1: Exercise

The solution of the exercises proposed in this document was given during class.

Exercise 1) [7 points]

Consider the following process composed by a tank of area $S = 2m^2$ and a reactor. The water from the tank is used to cool the reactor.

Tank

The water supposed to have a constant temperature of $T_r = 80^\circ C$ can be sent into the tank with a forced flow $q_i(t)$.

The level of the tank is controlled by the following

$$\frac{dh(t)}{dt} = \frac{1}{S}(q_i(t) - q_t(t))$$

The output of the tank has a flow $q_t(t)$ proportional to the level $h(t)$ of the tank.

$$q_t(t) = \sqrt{2gh(t)}$$

Reactor

The reactor is fed with a flow of hot water having a temperature $T_r = 80^\circ C$ with a given flow of $q_r(t)$

The water in the reactor is used to cool the reactor whose temperature (T) is given by the following

$$\frac{dT(t)}{dt} = 0.3(q_t(t)T_t - T(t) + q_r(t)T_r)$$

We are interested in the temperature $T(t)$ of the reactor.

1. Determine the dynamic equation that define the systems
2. Classify the system
3. Find an equilibrium point such that the reactor temperature is $40^\circ C$ considering a constant flow $q_r(t) = 10$
4. If the equilibrium point exists, determine, if possible, its stability.

Exercise 2) [5 points]

Given the following LTI System

$$\begin{cases} \dot{x}_2(t) = x_1(t) - 3x_2(t) \\ \dot{x}_1(t) = -2x_1(t) + 0.5u(t) \\ y_1(t) = x_1(t) + u(t) \\ y_2(t) = x_2(t) \end{cases}$$

1. Define the transfer function.
2. Is the transfer function a solid approximation of the system? Motivate the answer.

Exercise 3) [7 points]

Consider the following systems:

$$G_4(s) = \frac{100}{s^2 + 5s + 25}$$

and recalling the following:

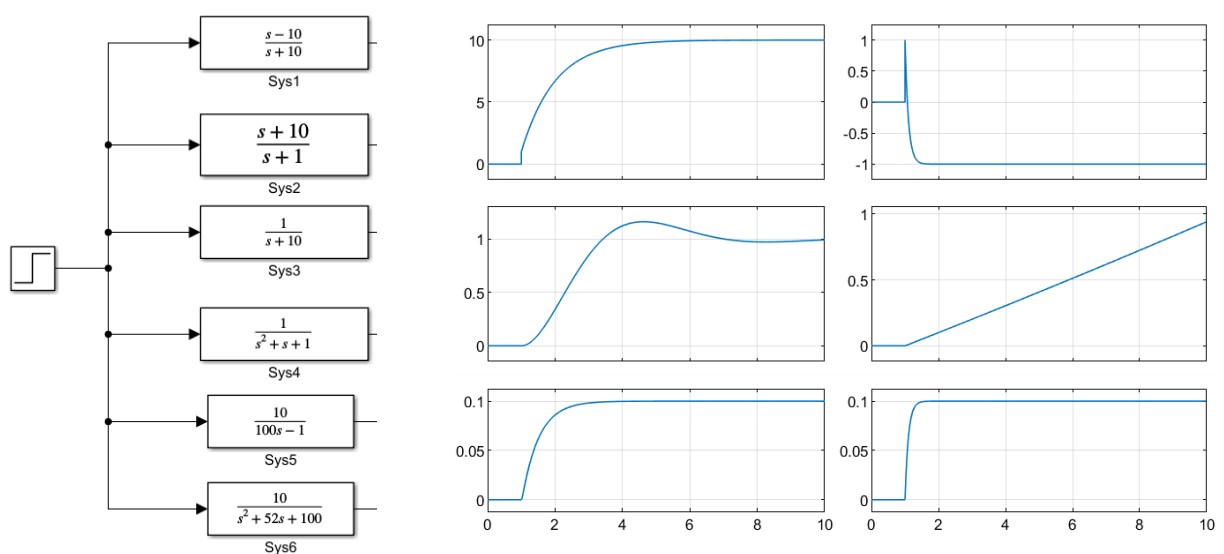
<i>1st Order Strictly Proper</i>				<i>1st Order Proper</i>	<i>2nd Order: complex pole</i>	
T_s	T_r	T_{a5}	T_{a1}	$T_{a\varepsilon}$	$S\%$	$T_{a\varepsilon}$
$\approx 2.2T$	$\approx 0.7T$	$\approx 3T$	$\approx 4.7T$	$T \ln \frac{ 1-\alpha }{0.01\varepsilon}$	$100e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$	$-\frac{\ln 0.01\varepsilon}{\xi\omega_n}$

determine the following indexes from a unitary step response

- Equilibrium value
- Settling time at 5%
- Overshooting percentage

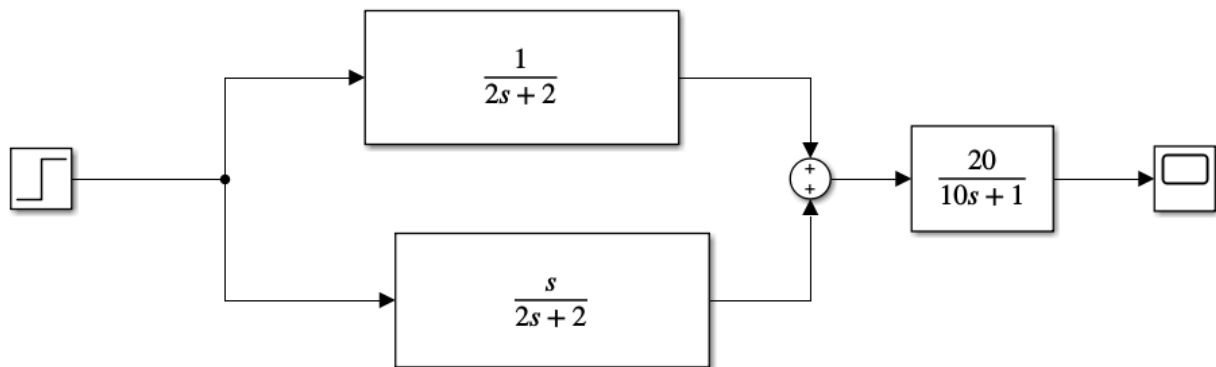
Exercise 4) [6 point]

Label each step response with the correct system



Exercise 5) [7points]

Given the system



1. evaluate the response to the following signal:
 - a. $\sin(100t) + 3 \cos(t)$
2. Define the Bode Diagram

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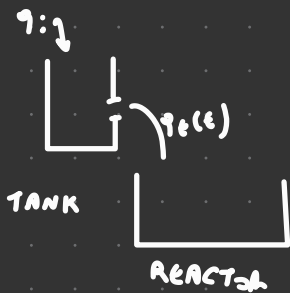
We are interested in the temperature $T(t)$ of the reactor.

- Determine the dynamic equation that define the systems
- Classify the system
- Find and equilibrium point such that the reactor temperature is $40\text{ }^{\circ}C$ considering a constant flow $q_r(t) = 10$
- If the equilibrium point exists, determine, if possible, its stability.

$$S = 2m^2$$

$$T_r = 80^{\circ}C$$

$$q_i(t)$$



$$\frac{dh(t)}{dt} = \frac{1}{S}(q_i(t) - q_t(t))$$

$$q_t(t) = \sqrt{2gh(t)} = q_1(t)$$

$$\frac{dT(t)}{dt} = 0.3(q_t(t)T_t - T(t) + q_r(t)T_r)$$

DYNAMIC EQUATION OF THE SYSTEM

$$\frac{dh(t)}{dt} = \dot{x}_1(t)$$

$$\dot{x}_1(t) = \frac{1}{S}(u_1(t) - \sqrt{2g}x_1(t))$$

$$\frac{dT(t)}{dt} = \dot{x}_2(t)$$

$$\dot{x}_2(t) = 0.3(\sqrt{2gx_1(t)}T_t - T(t) +$$

$$q_r(t)T_r)$$

$$u_2(t) = q_t(t) = \sqrt{2gh(t)}$$

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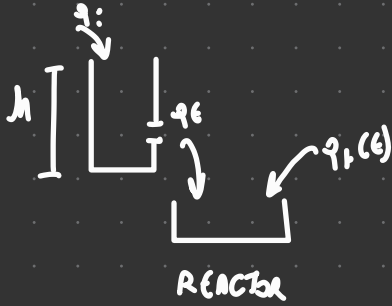
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$$\begin{cases} \dot{x}_1(t) = \frac{1}{3}(u_1(t) - \sqrt{2g}x_1(t)) \\ \dot{x}_2(t) = 0.3(\sqrt{2g}x_1(t)T_t - x_2(t) + u_2(t)T_r) \end{cases}$$

- 2nd ORDER ✓
- MISO ✓
- DYNAMIC ✓
- TIME INVARIANT ✓
- NON LINEAR ✓

• ~~PROPER~~ STRICLY PROPER / NOT IN OUTPUT FUNCTION

$$\begin{array}{l|l} S = 2m^2 & T_t = 80^\circ C \\ T_r = 80^\circ C & q_r(t) \\ \frac{dh(t)}{dt} = \frac{1}{3}(q_i - q_t(t)) & \frac{dT(t)}{dt} = 0.3(q_t(t)T_t - \boxed{T(t)} + q_r(t)T_r) \\ q_t(t) = \sqrt{2gh(t)} & \end{array}$$



$$\frac{dh(t)}{dt} = \dot{h}(t) = \frac{1}{3}(q_i(t) - q_t(t)) = \frac{1}{3}(q_i(t) - \sqrt{2gh(t)})$$

$$\frac{dT(t)}{dt} = \dot{T}(t) = 0.3(\sqrt{2gh(t)}T_t - T(t) + q_r(t)T_r)$$

- $u_1(t) = q_i(t)$ WATER FLOW IN TANK
- $u_2(t) = q_r(t)$ WATER FLOW IN REACTOR

$$eq: (\bar{x}_1, \bar{x}_2); (\bar{u}_1, \bar{u}_2) \quad T(t) = 40 \quad q_r(t) = 10$$

$$\begin{cases} \dot{x}_1(t) = \frac{1}{3}(u_1(t) - \sqrt{2g}x_1(t)) \\ \dot{x}_2(t) = 0.3(\sqrt{2g}x_1(t)T_t - x_2(t) + u_2(t)T_r) \end{cases} \longrightarrow \begin{cases} 0 = \frac{1}{3}(\bar{u}_1(t) - \sqrt{2g}\bar{x}_1(t)) \longrightarrow \bar{u}_1(t) - \sqrt{2g}\bar{x}_1(t) = 0 \\ 0 = 0.3(\sqrt{2g}\bar{x}_1(t)T_t - 40 + 10T_r) \longrightarrow \sqrt{2g}\bar{x}_1(t) = \frac{40 - 10T_r}{T_t} = \frac{40 - 10 \cdot 80}{80} \end{cases}$$
$$\bar{u}_1 - \sqrt{2g}\bar{x}_1(t) = \sqrt{2g}\bar{x}_1(t)T_t - 40 + 10T_r$$
$$\sqrt{2g}\bar{x}_1(t) = \frac{\bar{u}_1(t) + 40 - 10T_r}{T_t + 1}$$

$\sqrt{2g}\bar{x}_1(t) < -9.5$ UNSAT
STATE $x_1(t)$ CAN NOT BE NEG.